

Integrated CLIP-seq and Ribosome Profiling Analysis of LIN28A-Associated Translational Regulation

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Abstract

LIN28A is an RNA-binding protein involved in post-transcriptional regulation, including let-7 microRNA biogenesis and mRNA-level regulatory programs. In this project, CLIP-seq, ribosome profiling (RPF), and RNA-seq read counts were integrated to ask whether genes with stronger LIN28A-associated CLIP signal also show altered translation efficiency after *Lin28a* knockdown. A total of 20,553 genes passed the expression filter. For each gene, RNA-normalized CLIP enrichment and translation efficiency (TE) change between *siLin28a* and *siLuc* were calculated. Quadrant analysis identified 1,410 genes with high CLIP enrichment and increased TE, 881 genes with low CLIP enrichment and increased TE, 5,734 genes with low CLIP enrichment and decreased TE, and 4,294 genes with high CLIP enrichment and decreased TE. Gene biotype-based enrichment showed strong enrichment of processed pseudogenes in the high-CLIP/decreased-TE group ($\text{FDR} = 5.92\text{e-}102$) and enrichment of protein-coding genes in the low-CLIP/decreased-TE group ($\text{FDR} = 1.39\text{e-}72$). These results suggest that LIN28A-associated binding signal and translation response are related but not simply directional. Because the current enrichment analysis uses gene biotype categories as a GO-like proxy rather than a full Gene Ontology database, functional interpretation should be treated as exploratory.

Introduction

RNA-binding proteins regulate gene expression after transcription by influencing RNA processing, stability, localization, and translation. LIN28A is a developmentally important RNA-binding protein best known for repressing maturation of the let-7 microRNA family, but it has also been associated with broader mRNA regulatory functions [1,2]. Understanding LIN28A regulation therefore

requires more than identifying RNA binding alone: binding information should be interpreted together with RNA abundance and ribosome occupancy.

This project asks whether LIN28A-associated CLIP signal is linked to translational changes after **Lin28a** knockdown. CLIP-seq read counts were used as a proxy for RNA binding signal, while RPF/RNA ratios were used as a proxy for translation efficiency. These choices follow the general logic of CLIP-based RNA-protein interaction mapping and ribosome profiling-based translation measurement [3,4]. The analysis combines these assays into a gene-level matrix, computes CLIP enrichment and TE change, assigns genes to CLIP/TE response quadrants, and summarizes category-level enrichment patterns.

Methods

Data and Preprocessing

The analysis used the class dataset files `CLIP-35L33G.bam`, `RPF-siLuc.bam`, `RPF-siLin28a.bam`, `RNA-siLuc.bam`, `RNA-siLin28a.bam`, and `encode.gtf`. Gene-level annotation was derived from the GENCODE GTF file [6]. Reads were counted at the gene level for each assay using a `featureCounts`-style gene assignment strategy [5], then merged by `gene_id` to create a single integrated matrix containing CLIP, RPF, and RNA counts.

Genes were retained if the sum of RNA and RPF counts across the two conditions was at least 10. This filtering step retained 20,553 genes. Sample-level assigned read counts were 13,548,748 for CLIP, 12,925,056 for RPF siLuc, 10,688,552 for RPF siLin28a, 9,736,263 for RNA siLuc, and 12,321,224 for RNA siLin28a. The number of detected genes ranged from 18,055 to 20,312 across assays.

CLIP Enrichment and Translation Efficiency Metrics

CLIP enrichment was normalized by RNA abundance to reduce the effect of highly expressed genes:

$$\text{CLIP per RNA} = (\text{clip_count} + 1) / (\text{mean}(\text{RNA-siLuc}, \text{RNA-siLin28a}) + 1)$$

$$\text{CLIP enrichment log2} = \log_2(\text{CLIP per RNA} / \text{median}(\text{CLIP per RNA}))$$

Translation efficiency was computed as the RPF/RNA ratio in each condition, followed by a log2 fold change:

$$\text{TE_siLuc} = (\text{RPF-siLuc} + 1) / (\text{RNA-siLuc} + 1)$$

$$\text{TE_siLin28a} = (\text{RPF-siLin28a} + 1) / (\text{RNA-siLin28a} + 1)$$

$$\text{TE change log2} = \log_2(\text{TE_siLin28a} / \text{TE_siLuc})$$

Across all retained genes, the median CLIP enrichment was 0.000 and the mean was -0.0047, indicating that the CLIP metric was centered by construction. The median TE change was -0.778 and the mean was -0.728, showing an overall shift toward lower TE in the **siLin28a** condition. Spearman correlation between

CLIP enrichment and TE change was 0.178 ($p = 6.94e-146$), indicating a weak but statistically detectable positive association.

Quadrant Definition

Genes were classified into CLIP/TE response groups using fixed thresholds:

- **Q1_clip_high_te_up**: CLIP enrichment > 0.5 and TE change > 0.3
- **Q2_clip_low_te_up**: CLIP enrichment ≤ -0.5 and TE change > 0.3
- **Q3_clip_low_te_down**: CLIP enrichment ≤ -0.5 and TE change ≤ -0.3
- **Q4_clip_high_te_down**: CLIP enrichment > 0.5 and TE change ≤ -0.3
- **other**: genes not assigned to the four threshold-defined quadrants

Enrichment Analysis

The current project does not use a full external Gene Ontology gene set database [7]. Instead, it performs a GO-like enrichment analysis using **gene_type** categories from the annotation file. For each quadrant, Fisher's exact test was used to test whether a gene biotype category was overrepresented relative to the full retained gene universe. P-values were adjusted using the Benjamini-Hochberg false discovery rate procedure [8]. The results should therefore be interpreted as gene biotype/category enrichment, not as definitive biological-process GO enrichment.

Results

Generated Figures

The extended figure set was generated in `03_results/figures/`:

Gene-Level CLIP and TE Patterns

Among the retained genes, 1,410 were classified as high CLIP/increased TE, 881 as low CLIP/increased TE, 5,734 as low CLIP/decreased TE, and 4,294 as high CLIP/decreased TE. The remaining 8,234 genes did not meet the quadrant thresholds. The two decreased-TE quadrants together contained 10,028 genes, indicating that TE reduction after **Lin28a** knockdown was the dominant direction in this dataset.

Group-level summaries showed that **Q1_clip_high_te_up** had mean CLIP enrichment of 1.85 and mean TE change of 1.12. In contrast, **Q4_clip_high_te_down** had mean CLIP enrichment of 1.55 but mean TE change of -1.27. Thus, strong CLIP signal was not uniformly associated with increased translation after knockdown. This pattern suggests that LIN28A-associated binding may mark multiple regulatory contexts rather than a single monotonic relationship between binding and TE response.

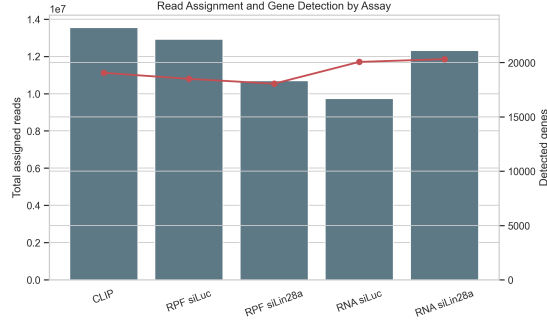


Figure 1: Sample Count QC. The figure summarizes the basic read-count quality control for the five assays used in the project: CLIP, RPF siLuc, RPF siLin28a, RNA siLuc, and RNA siLin28a. The bar plot shows the total number of assigned reads for each sample, while the line plot shows the number of genes with at least one detected read. This figure is used to confirm that each assay has enough assigned reads and that gene detection is broadly comparable across samples. The RNA samples detected the largest number of genes, while the RPF and CLIP samples still showed sufficient coverage for downstream gene-level analysis.

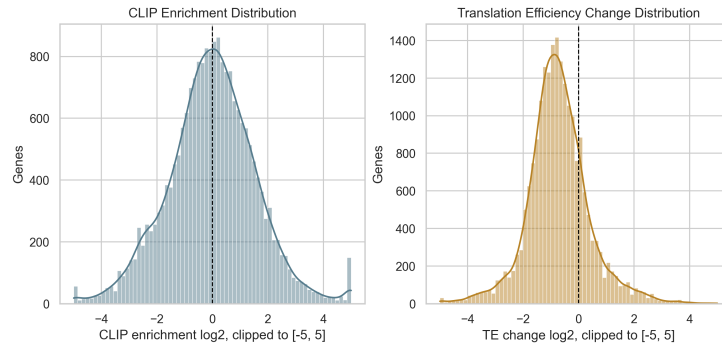


Figure 2: Metric distributions. It shows the distributions of the two main gene-level metrics used in the analysis: CLIP enrichment and translation efficiency change. The CLIP enrichment distribution is centered near zero because the metric was normalized by the median CLIP-per-RNA value. The TE change distribution is shifted toward negative values, indicating that many genes show reduced translation efficiency in the ‘siLin28a’ condition compared with ‘siLuc’. This figure supports the observation that TE decrease is a dominant genome-wide trend after ‘Lin28a’ knockdown.

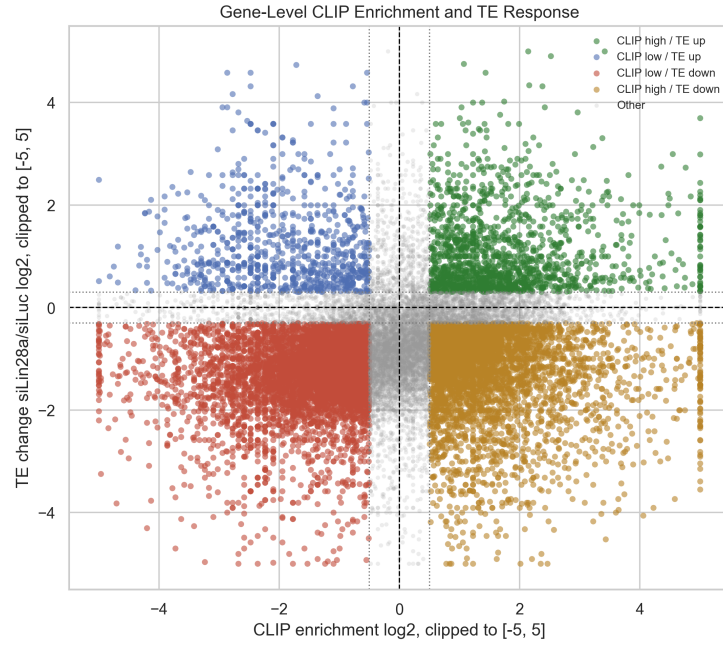


Figure 3: Gene-level CLIP enrichment versus TE change scatter plot. Each point represents one gene. The x-axis shows RNA-normalized CLIP enrichment, and the y-axis shows log2 TE change between ‘siLin28a’ and ‘siLuc’. Dashed and dotted guide lines indicate the thresholds used to define the four CLIP/TE quadrants. Genes in the upper-right region have high CLIP enrichment and increased TE, while genes in the lower-right region have high CLIP enrichment but decreased TE. This figure shows that CLIP-enriched genes are distributed across both TE-increased and TE-decreased regions, suggesting that LIN28A-associated binding is not linked to one simple direction of translation change.

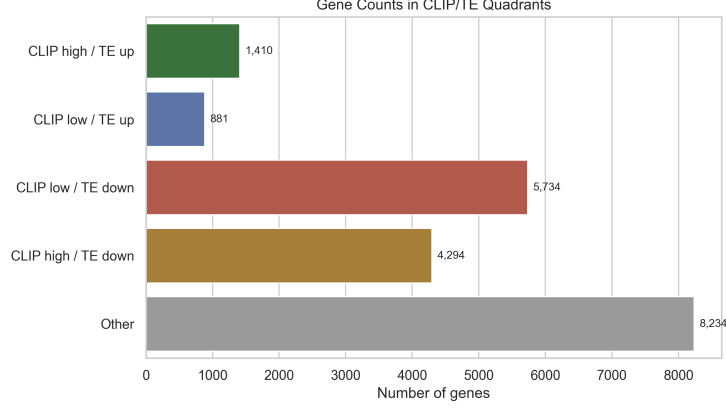


Figure 4: Gene counts in CLIP/TE quadrants. Bar plot shows how many genes were assigned to each CLIP/TE response group. The largest threshold-defined group is ‘CLIP low / TE down’, followed by ‘CLIP high / TE down’. The two TE-down groups together contain more genes than the two TE-up groups, supporting the conclusion that TE reduction is the major response pattern in this dataset. The ‘Other’ category includes genes whose CLIP enrichment or TE change did not pass the quadrant thresholds.

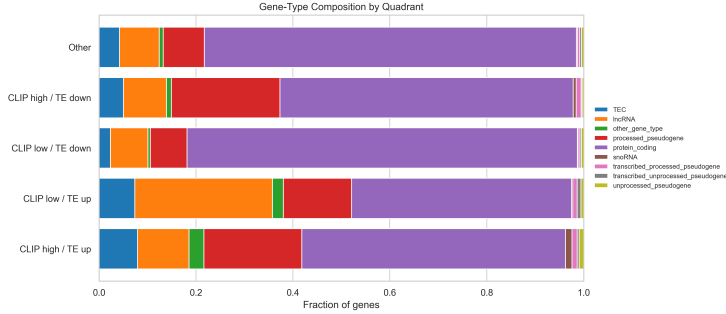


Figure 5: Gene-type composition by quadrant. It shows the fraction of major gene biotypes within each CLIP/TE quadrant. It helps determine whether certain response groups are dominated by specific annotation categories such as protein-coding genes, lncRNAs, pseudogenes, or small RNA categories. The plot shows that the quadrants differ in gene-type composition, which is important because some apparent CLIP/TE patterns may reflect annotation or mapping properties of particular gene biotypes rather than only direct biological regulation.

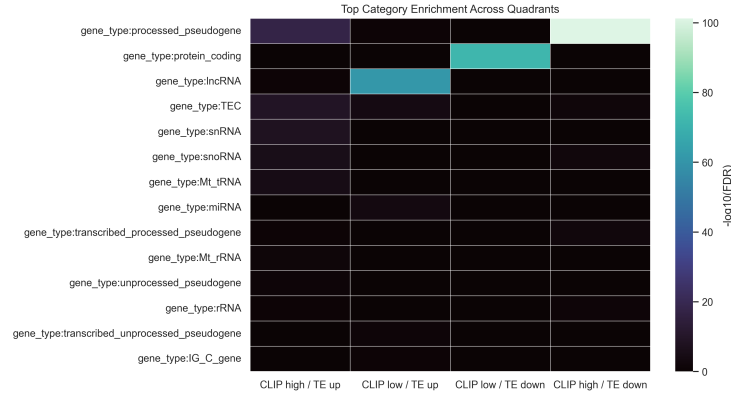


Figure 6: Enrichment heatmap. It summarizes the most significant gene biotype enrichment results across the CLIP/TE quadrants. Darker color indicates stronger enrichment, measured as ‘ $-\log_{10}(\text{FDR})$ ’. The heatmap makes it easier to compare which categories are enriched in each response group. For example, processed pseudogenes show strong enrichment in the ‘CLIP high / TE down’ group, while protein-coding genes are strongly enriched in the ‘CLIP low / TE down’ group.

Gene Biotype Enrichment

The strongest enrichment result was observed for **gene_type:processed_pseudogene** in the high-CLIP/decreased-TE group ($\text{FDR} = 5.92\text{e-}102$, overlap = 963). The low-CLIP/decreased-TE group was strongly enriched for **gene_type:protein_coding** genes ($\text{FDR} = 1.39\text{e-}72$, overlap = 4,618). The low-CLIP/increased-TE group was enriched for **gene_type:lncRNA** ($\text{FDR} = 2.17\text{e-}61$, overlap = 250). In the high-CLIP/increased-TE group, processed pseudogene, TEC, snRNA, snoRNA, and mitochondrial tRNA categories appeared among the top enriched categories.

Mean values by major biotype were also consistent with these enrichment patterns. Protein-coding genes had mean CLIP enrichment of -0.123 and mean TE change of -0.761. lncRNAs had mean CLIP enrichment of -0.406 and mean TE change of -0.490. Processed pseudogenes had higher mean CLIP enrichment (0.769) but lower mean TE change (-0.860), matching their strong representation in the high-CLIP/decreased-TE quadrant.

Discussion

This analysis integrates CLIP-seq, ribosome profiling, and RNA-seq counts to summarize the relationship between LIN28A-associated binding signal and translational response after **Lin28a** knockdown. The overall TE distribution shifted downward in the knockdown condition, while the relationship between

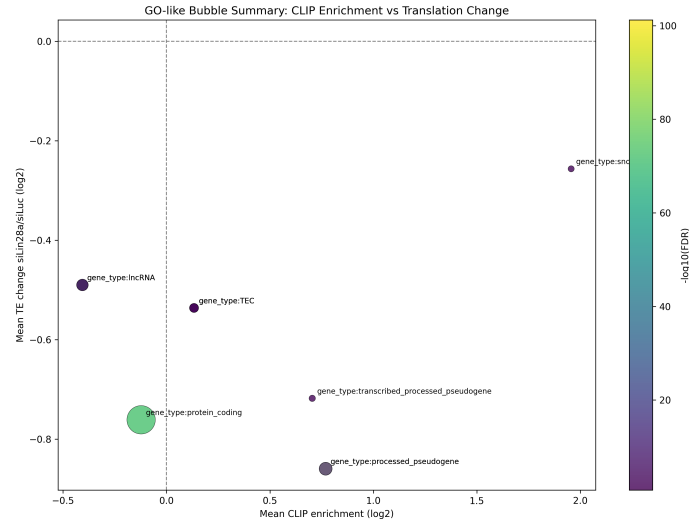


Figure 7: GO-like CLIP/RPF bubble summary plot. It summarizes enriched gene categories in the same coordinate system used for the main analysis. The x-axis represents mean CLIP enrichment for genes in a category, and the y-axis represents mean TE change. Bubble size represents the number of genes in the category, and bubble color represents enrichment significance using ‘ $-\log_{10}(\text{FDR})$ ’. Categories on the right have stronger average CLIP enrichment, while categories above zero show increased average TE after ‘Lin28a’ knockdown. Because the current enrichment uses gene biotypes rather than true GO biological process terms, this plot should be interpreted as a GO-like category summary rather than a final pathway enrichment result.

CLIP enrichment and TE change was weakly positive. This indicates that CLIP signal alone does not fully explain the translation response, although it may contribute to a broader regulatory pattern.

The large high-CLIP/decreased-TE group is especially notable because it argues against a simple model in which stronger LIN28A-associated binding always corresponds to increased TE after knockdown. However, the enrichment of processed pseudogenes, snRNAs, snoRNAs, and rRNAs should be interpreted cautiously. These categories can be affected by multi-mapping, repetitive sequence content, annotation structure, or gene-level counting artifacts. The protein-coding enrichment in the low-CLIP/decreased-TE group may reflect a broader global translation shift or RNA abundance effect rather than direct LIN28A binding.

The main limitation is that the enrichment analysis currently uses gene biotype categories instead of true Gene Ontology biological process terms. Therefore, the bubble plot and heatmap are best described as GO-like category summaries. A stronger follow-up analysis would map mouse Ensembl gene IDs or gene symbols to a real GO database and perform biological process enrichment on each quadrant. Additional improvements would include replicate-aware differential expression and differential translation analysis, peak-level CLIP quantification, and region-specific binding analysis across CDS, UTR, intronic, and noncoding regions.

Conclusion

The project provides an exploratory gene-level framework for integrating LIN28A-associated CLIP signal with translation efficiency changes after *Lin28a* knockdown. The results show a broad decrease in TE, a weak association between CLIP enrichment and TE change, and strong gene-biotype-specific patterns across CLIP/TE quadrants. The current findings are useful for hypothesis generation, but a full GO enrichment analysis and replicate-aware statistical modeling would be needed for publication-grade biological conclusions.

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